Software Requirement Specification

for

DieHard Collector

Version 1.0 Approved

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I. Revision History

Name	Date	Reason For Change	Version
Initial Version (Func. + Non-Func. Requirements)	2/27/2026	Initial for Deliverable 1	1.0
First revision (adds remainder of document)	3/12/2026	Deliverable 2 requirements	1.1
Second Revision post D2	4/4/2026	Deliverable 2 Update	1.2
Third Revision during deliverable 4, based on feedback from deliverable 2	4/16/2026	Deliverable 4 Update	1.3

1. Introduction

1.1 Purpose

DieCard Collector is an app designed to help trading card collectors organize their collection efficiently on the computer. The app gives users a way to add new cards to a collection and an intuitive search system that supports advanced specification (e.g., card set ID). The app supports existing trading card databases, allowing users to select their desired card and place it into one large collection or multiple smaller ones. The following document describes the software specifications needed to create this application.

1.2 Document Conventions

This document was created based on the IEEE template for System Requirements Specification (SRS) Documents.

Additional conventions used are as follows:

Convention	Description
DieCard	Application Name Group/Company Name
Document	In reference to this SRS Document
Application	In reference to the DieCard web application
React	Front-end JavaScript and TypeScript-supported website development library

1.3 Intended Audience and Reading Suggestions

This Document is intended for the Application's developers, testers, project managers, and Application users to understand the goals and the Application's intended purpose. This Document includes an overall description of the project, the external requirements, and the functional and non-functional requirements for the Application.

Non-developer users are suggested to read **sections 1, 2, and 3** in their entirety to comprehend the Application's intentions, structure, and essence.

Testers are suggested to read the following sections:

- **1. "Introduction"**, for an overview of the project.
- **3.2-3.4 "External Interface Requirements"**, for determining if the tester can run the Application.
- **4. "System Features"** and **5. "Other Nonfunctional Requirements"**, for testing cases.

Developers and project managers are expected to read this document in its entirety.

1.4 Product Scope

DieCard is a React-based web application where users input physical trading cards they own, or wish to own, into a digital collection. This Application is targeted towards dedicated trading card collectors who wish to amplify their collection experience with an adaptable, understandable tool. This Application promises an intuitive, rewarding collection experience.

1.5 References

DieCard Collector GitHub Repository: https://github.com/COS420-S2025/Team-C

2. Overall Description

2.1 Product Perspective

DieCard Collector is a new open-source application. It is intended for all trading card collectors who are dissatisfied with their current method of organizing their collections. The app relates to physical trading cards as an extension and a way to organize, but is not affiliated with any trading card companies. Existing solutions such as Rare Candy, Moxfield, and Archidekt are unintuitive and uncustomizable. DieCard seeks to fill that gap.

2.2 Product Functions

Account Management:

- The system will allow users to create accounts with their email and a password.
- The system will allow users to create accounts with their Google Login.
- The system will allow users to change their bound email or password.
- The system will allow users to log in to their created accounts using their credentials.

Collection Management:

- The system will allow users to create "collections" in which they can organize their cards.

- The system will allow users to upload cards to place in collections.
- The system will allow users to search & filter through their cards and collections.

Social:

- The system will allow users to create public collections, which other users can see.

2.3 User Classes and Characteristics

The user classes defined in the use cases are: collectors, buyers, and sellers. Sellers and buyers extend collectors, but also intend to use the application to trade, buy, or sell cards with others. The highest priority consumers are those who are experienced collectors, since they are more interested in the high customizability than a newer collector, although DieCard is intended for all collectors.

2.4 Operating Environment

DieCard collector is designed to operate on mobile and computer web browsers. It will function on any operating system as long as it supports the up-to-date version of Google Chrome, Mozilla Firefox, Microsoft Edge, or any other browser that supports JavaScript.

2.5 Design and Implementation Constraints

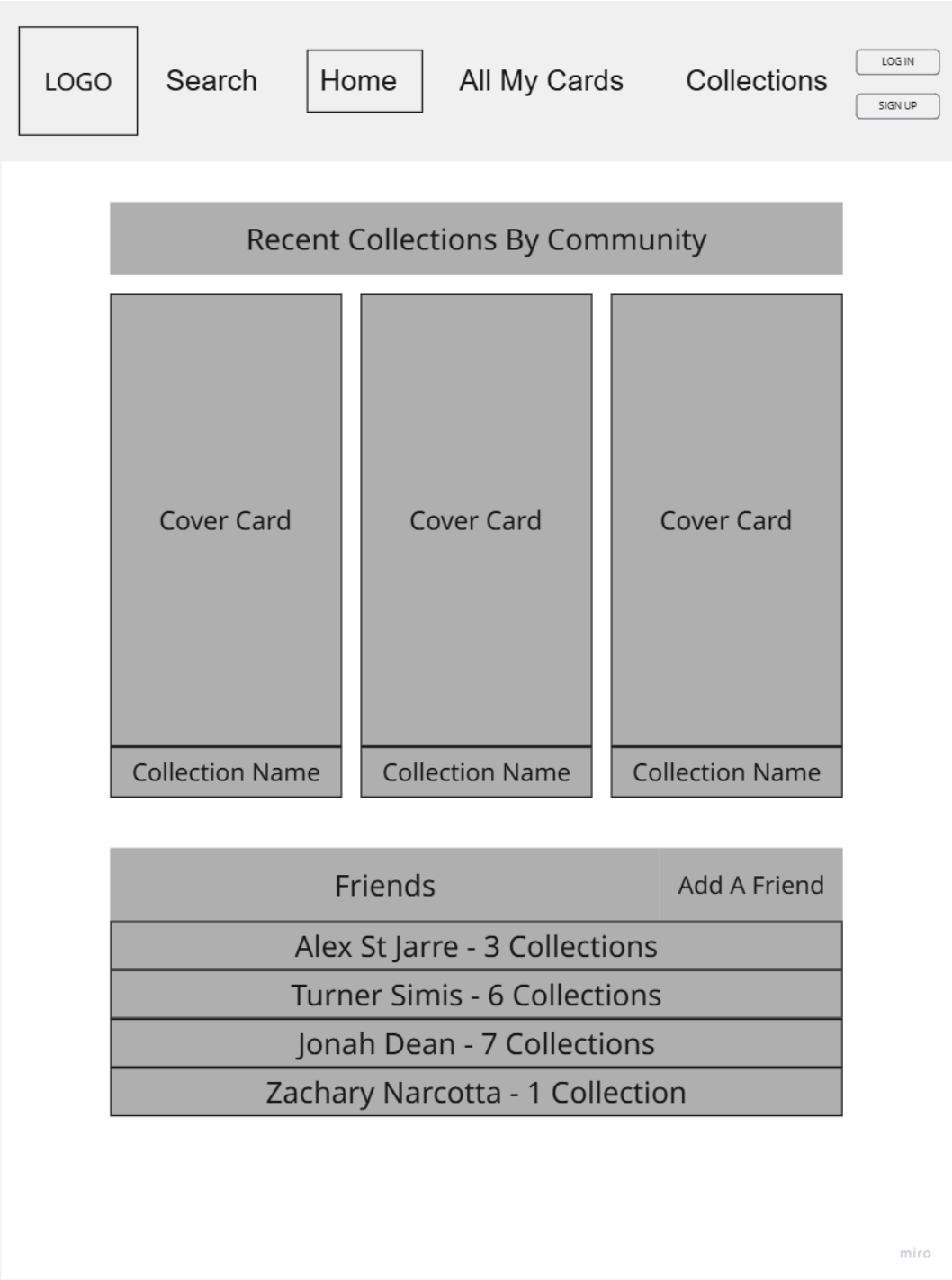
The DieCard project has a short development period: from Feb. 5, 2026, to May. 7, 2026—just over 3 months. As such, some features may not be available or may be erroneous at launch. Additionally, DieCard uses the *TCGDex* database for its card search, which may not be exhaustive.

2.6 Assumptions and Dependencies

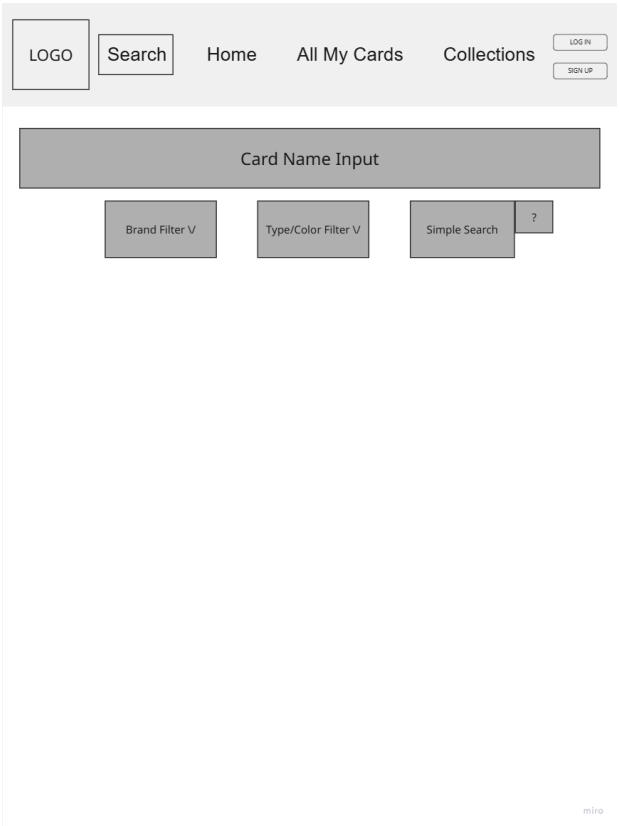
DieCard depends on the user running a browser with JavaScript enabled. It also depends on the backend server to stay active and running for the cards to update properly.

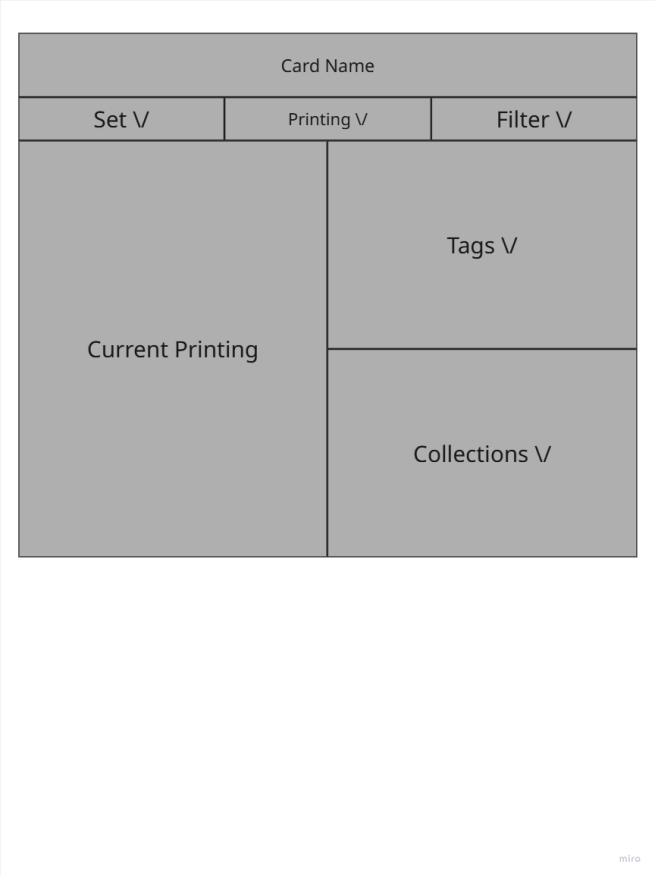
3. External Interface Requirements

3.1 User Interfaces

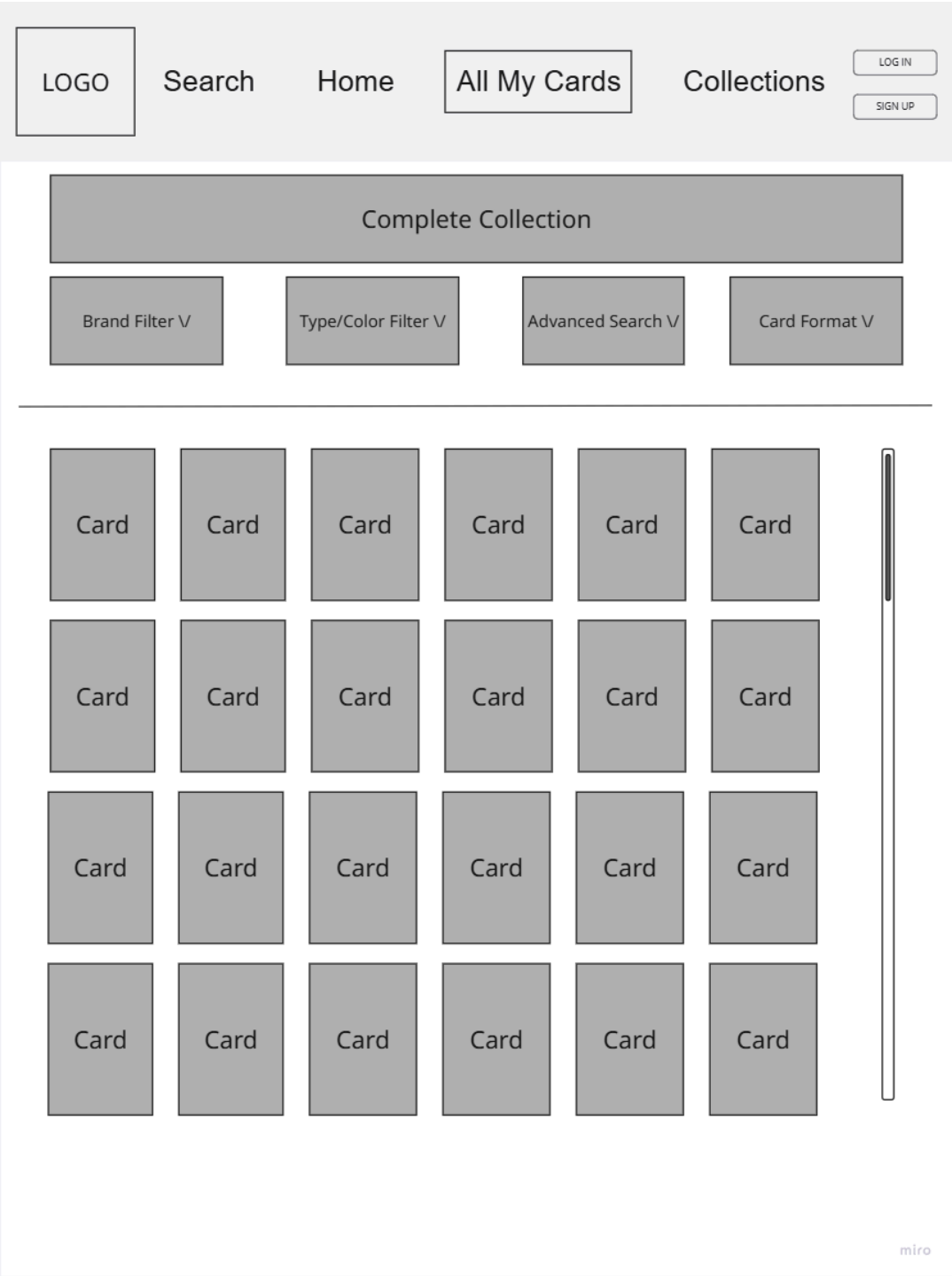


This is an example of what our login page will look like. Above is a quick introduction to our app and directions for the page. On the main page, there will be collections uploaded by other users that you have friended in the app. You will also be able to add friends on this page by entering their username. There is a login button in the top right to enter information in the form of a username/email and password to create a new account for our site. Our UI will be designed to work on Desktop with the user's preferred browser.

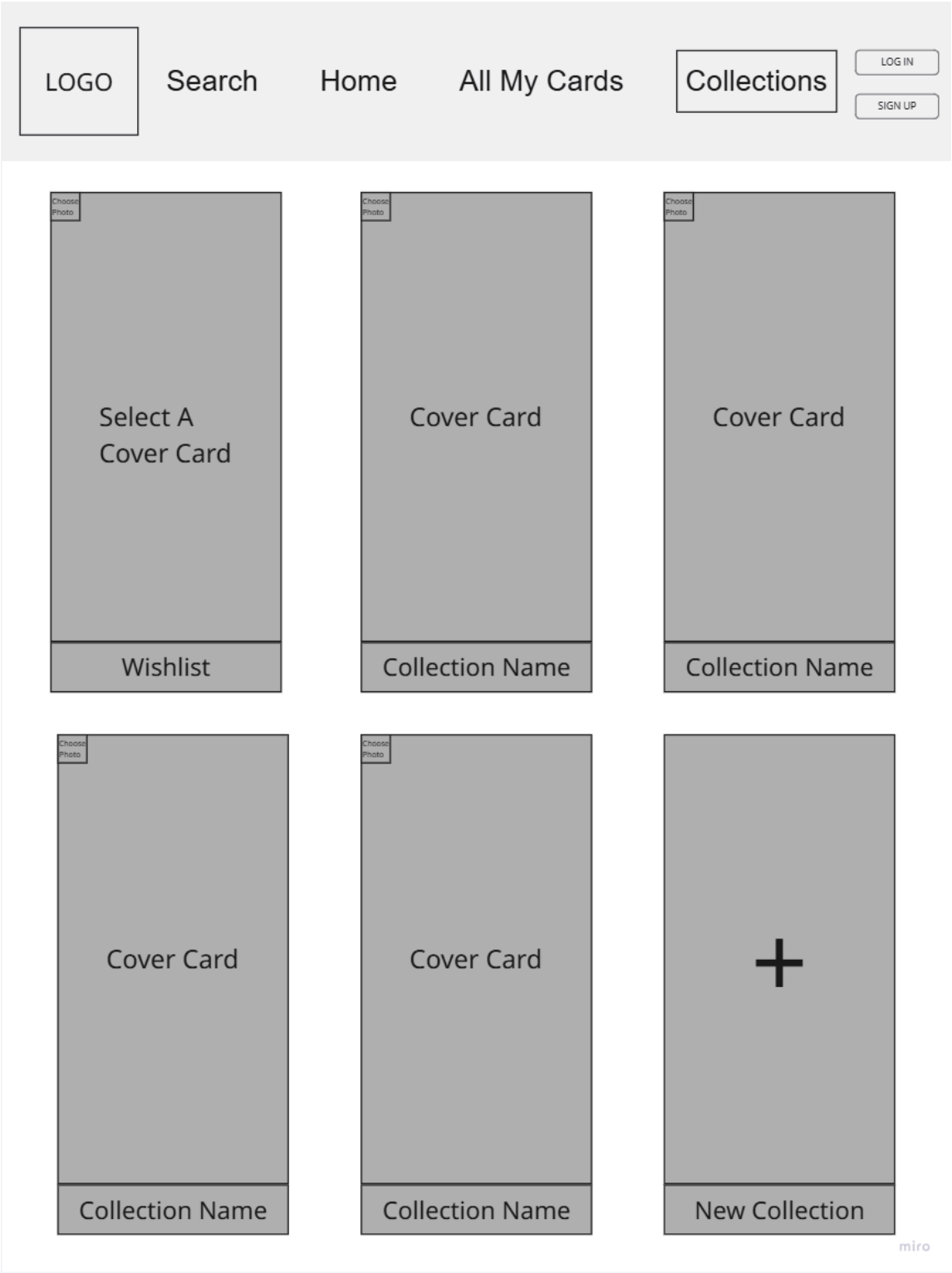




This is a basic look at what our card search page will look like. On the left is the layout of the page, and on the right is a card once it has been searched. You will be able to choose what page you want from the available tabs at the top of the page. You are able to input a card name into the search bar and view your desired card if it is in the publicly available databases. From there you will be able to add it to your collection, or keep looking for more cards. There are a few options to filter the search results. You will be able to select the brand of cards that you want to search from, as well as different defining attributes and printings that the cards have.



This is a basic look at the user’s card collection page that is supposed to be the digital representation of their physical collection. This page will have similar search and filtering features as the search page. The cards will then be displayed in a grid pattern to the user. When a card is clicked, the system will display a larger png of the card as well as detailing its attributes and allowing the user to add it to a collection. The user will have the option of entering a card's name and adding it to the collection to streamline the process of adding all of your cards to the app without any other information of the card being stored.



This is a basic look of the collections page. The user will have a button available to them to create a new collection, which will create the opportunity for them to add a card to this specific user created collection. The user can select a card for the cover of the collection. All of the collections are stored in this page and displayed in a grid layout.

4. System Features

4.1 Login

4.1.1 Description:

The system shall allow users to create an account to check their collection status, add cards, and create new collections.

Priority: High

4.1.2 Response Sequences

4.1.2A Sign Up

The user selects “Sign Up.”
The system displays the Sign Up page.
The user enters:
- Username
- Email
- Password

The user submits the form.
The system validates information
The user completes 2FA verification.
The system confirms successful account creation.

4.1.2B Login

The user selects “Login.”
The system displays a Login page.
The user enters:
- Username or Email
- Password

The user submits a login form.
The system validates credentials.
The system prompts for 2FA verification.
The user enters 2FA code.
The system logs the user in.
The system opens the Home Page.

4.1.2C Guest Access

The user selects “Continue as Guest.”
The system grants limited access.
The system redirects to the Home Page.
The system restricts buying/selling features.

4.2.1D Logout

The user selects “Logout.”
The system ends the session.
The system redirects to the Login/Home page.
The system clears session tokens.

4.1.3 Functional Requirements

REQ-1: The system shall allow a user the option to create an account for buying/selling cards.
REQ-2: The system shall allow a user to create an account with a unique username and password with the option of using an email address.
REQ-3: The system shall display success when the user account is created and proceed while keeping the user signed in.
REQ-4: The system shall not require the user to activate an account to utilize the collection features (i.e., “guest access”).
REQ-5: The system shall allow the user to log out of their account once signed in

4.2 Collection

4.2.1 Description

The system shall allow users to create multiple collections that have specific cards that belong in the list as well as a wanted list for that collection to show missing cards for that collection

Priority: Medium

4.2.2 Response Sequences

4.2.2A Create New Collection

The user navigates to “Collections” tab.
The system displays a list of collections.
The user selects “Create Collection.”
The system prompts for:
- Collection name
- Category
- Description

The user submits form.
The system creates the collection.

The system displays the new collection in the list.

4.2.2B Delete Card

The user selects a card within the collection.
The user selects “Remove.”
The system confirms deletion.
The user confirms.
The system removes card(s) from collection.

4.2.2C Delete a collection

The user selects a collection.
The user selects “Delete.”
The system confirms deletion.
The user confirms.
The system deletes the collection.
The system updates collection list.

4.2.2D View Card Details

The user selects a card.
The system retrieves card attributes.
System displays card information

4.2.3 Functional Requirements

REQ-1: The system shall allow users to create unique lists of cards within a tab called collections.
REQ-2: The system shall allow users to organize their collections using different categories and descriptions
REQ-3: The system shall create a “master” collection, which cannot be deleted, containing all user-added cards.
REQ-4: The system shall facilitate the addition and deletion of individual cards or entire collections.
REQ-5:The system shall display the selected card’s attributes and details when a user selects a card

4.3 Card Search

4.3.1 Description

The system shall allow users to search for any cards they want to add into a collection using a primary search area of the website or quickly in the card collection area

Priority: Medium

4.3.2 Response Sequences

4.3.2A Search

The user enters the card name in the search bar.
The system sends request to the Card Database API.
API returns matching results.
The system displays results.
The user selects a card.
The system displays card details.
The user selects “Add to Collection.”
The system updates selected collections.

4.3.2B Advanced Search

The user selects “Advanced Search.”
The system displays filters
The user selects filters.
The user submits a search.
The system queries API with filters.
The system displays filtered results.

4.3.3 Functional Requirements

REQ-1: The system shall allow users to search for cards by name and/or feature(s).
REQ-2: The system shall allow users to view cards not in their “master” collection
REQ-3: The system shall provide an advanced search feature that allows users to filter cards based on specific attributes.
REQ-4: The system shall allow users to add and remove cards which have been searched.

4.4 Social Feature

4.4.1 Description

This feature is the application’s sharing of collections and viewing other users’ collections.

Priority: Medium

4.4.2 Response Sequences

4.4.2A Set Collection to Public/Private

The user opens a collection.
The user selects “Privacy Settings.”
The user selects:
    Public
    Private
The user saves changes.
The system updates privacy status.

4.4.2B Search for Another User
The user enters another user’s username in search.
The system searches the user database.
The system displays matching users.
The user selects a user.
The system checks collection privacy:
    If Public - Display collection.
    If Private - Display “Collection is Private.”

4.4.3 Functional Requirements

REQ-1: The system shall allow users to set their collection to “public” or “private,” so that a user can control whether other users can see their collection.
REQ-2: The system shall allow users to view another user’s “public” collection by searching their username, so that users can see other users’ collections.
REQ-3: The system shall deny users from viewing another user’s “private” collection, so that users’ privacy selections are respected.
REQ-4: The system shall allow users to change their existing “public” collection to “private,” and vice versa, so that users may update their privacy selections.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NON-REQ-1: The system shall load the site in no more than 5 seconds, at least 99% of the time
NON-REQ-2: The system shall let users transfer between pages on a website in under 5 seconds, at least 99% of the time.
NON-REQ-3: The system shall add or remove a card from the user’s collection in no more than 2000ms, at least 99% of the time.
NON-REQ-4:The system shall support at least 20 concurrent users.
NON-REQ-5:The system shall return search results within 3 seconds at least 95% of the time.

NON-REQ-6: The backend system shall not exceed 5 minutes to recover after failure.

5.2 Safety Requirements

NON-REQ-1 The system shall ensure that user account data is transferred to the database, provided a stable user internet connection and valid user account creation.

5.3 Security Requirements

NON-REQ-1: The system shall require for all accounts passwords containing 8 or more characters, with at least one number and at least one special character (non-alphanumeric).
NON-REQ-2: The system must ensure that user account information (i.e., email, password) is securely encrypted and transmitted.
NON-REQ–3: The systems shall not display personal user information (See glossary) to other users.

5.4 Software Quality Attributes

NON-REQ-1: The system shall prompt users with a forgotten password prompt if more than 2 consecutive login attempts are unsuccessful.

6. Other Requirements

Appendix A: Glossary

DieCard: The company name

Personal User Information: Includes but is not limited to email address, password, google account, user id.

Appendix B: Backups

Backups are kept on Google Drive.